

Programming Methodology - SE1012

Lab 05 – Logical Expressions



Access Lab 5 via GitHub:

<https://classroom.github.com/a/bRduX7Y2>

Outline of Today's Lab



- Relational and Equality Operators
- Logical Operators
- Operator Precedence
- Short-Circuit Evaluation
- Question 1: Expression Evaluation with Short-Circuit Logic
- Question 2: Relational Condition Checks
- Question 3: Access Control System Logic
- Question 4: Test Pass/Fail Condition

Relational and Equality Operators

Operator	Meaning	Type
<	less than	relational
>	greater than	relational
<=	less than or equal to	relational
>=	greater than or equal to	relational
==	equal to	equality
!=	not equal to	equality

Sample Conditions

x	power	MAX_POW	y	item	MIN_ITEM	mom_or_dad	num	SENTINEL
-5	1024	1024	7	1.5	-999.0	'M'	999	999

Operator	Condition	English Meaning	Value
<code><=</code>	<code>x <= 0</code>	x less than or equal to 0	1 (true)
<code><</code>	<code>power < MAX_POW</code>	power less than MAX_POW	0 (false)
<code>>=</code>	<code>x >= y</code>	x greater than or equal to y	0 (false)
<code>></code>	<code>item > MIN_ITEM</code>	item greater than MIN_ITEM	1 (true)
<code>==</code>	<code>mom_or_dad == 'M'</code>	mom_or_dad equal to 'M'	1 (true)
<code>!=</code>	<code>num != SENTINEL</code>	num not equal to SENTINEL	0 (false)

Logical Operators

The `&&` Operator (and)

operand1	operand2	operand1 <code>&&</code> operand2
nonzero (true)	nonzero (true)	1 (true)
nonzero (true)	0 (false)	0 (false)
0 (false)	nonzero (true)	0 (false)
0 (false)	0 (false)	0 (false)

The `||` Operator (or)

operand1	operand2	operand1 <code> </code> operand2
nonzero (true)	nonzero (true)	1 (true)
nonzero (true)	0 (false)	1 (true)
0 (false)	nonzero (true)	1 (true)
0 (false)	0 (false)	0 (false)

The `!` Operator (not)

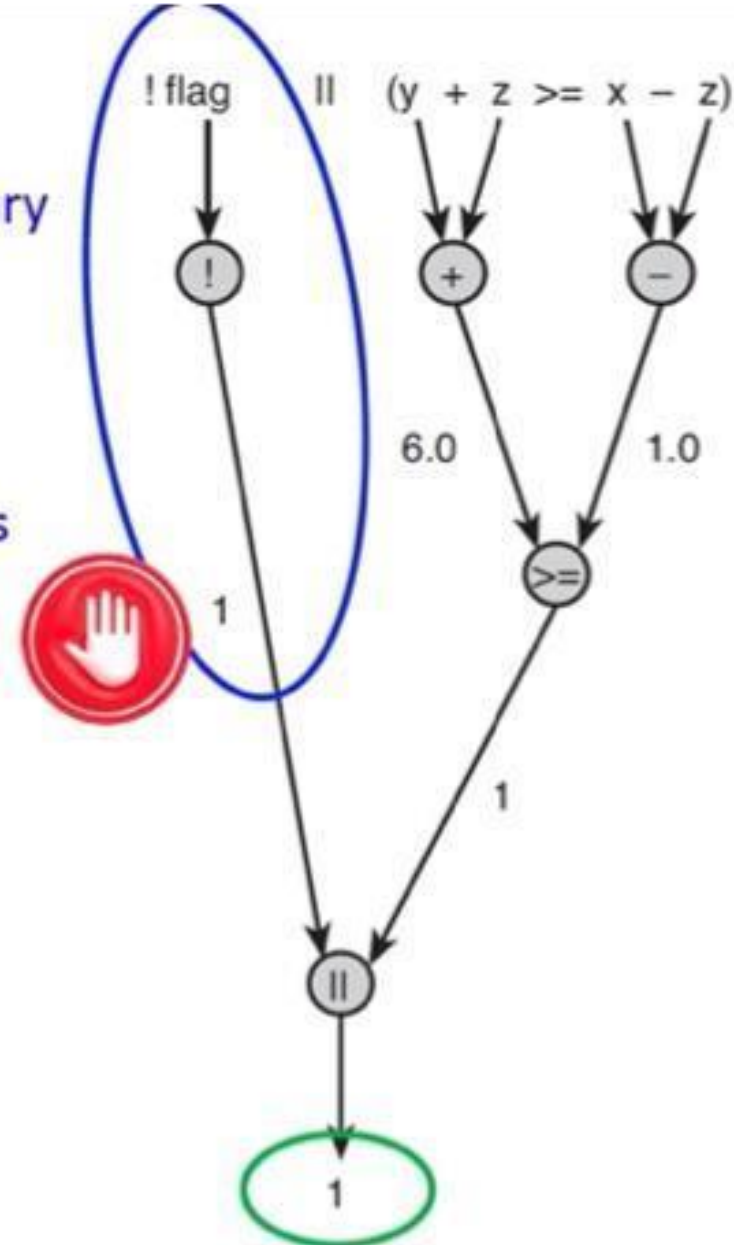
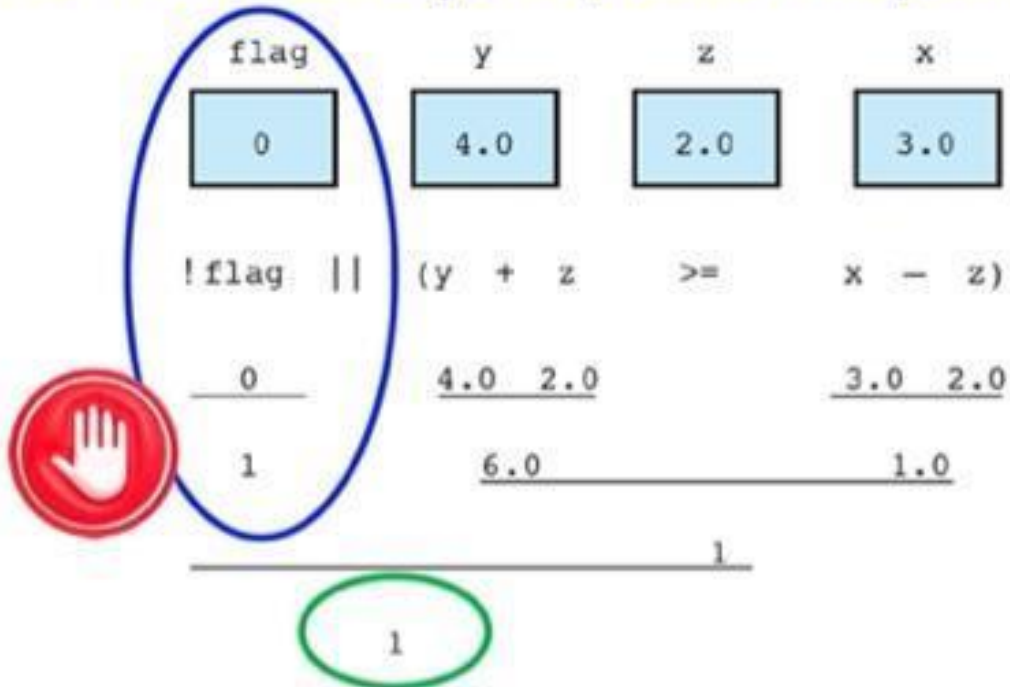
operand1	<code>!operand1</code>
nonzero (true)	0 (false)
0 (false)	1 (true)

Operator Precedence

Operator	Precedence
function calls / parentheses ()	highest
! + - & (unary operators)	
* / %	
+ -	
< <= >= >	
== !=	
&& (AND)	
(OR)	
=	lowest

Short-Circuit Evaluation

- Logical expressions are **evaluated only as far as necessary**
- **&& (AND)** stops if the first operand is false
- **|| (OR)** stops if the first operand is true
- Prevents unnecessary computation and potential errors



Question 1: Expression Evaluation with Short-Circuit Logic

Evaluate each of the following expressions if `a` is 6, `b` is 9, `c` is 14, and `flag` is 1. Which parts of these expressions are not evaluated due to short-circuit evaluation?

1. `c == a + b || !flag`
2. `a != 7 && flag || c >= 6`
3. `!(b <= 12) && a % 2 == 0`
4. `!(a > 5 || c < a + b)`

Question 2: Relational Condition Checks

Write an expression to test for each of the following relationships:

1. `age` is from 18 to 21 inclusive.
2. `water` is less than 1.5 and also greater than 0.1.
3. `year` is divisible by 4. (Hint: Use `%`.)
4. `speed` is not greater than 55.
5. `y` is greater than `x` and less than `z`.
6. `w` is either equal to 6 or not greater than 3.

Question 3: Access Control System Logic

Write a C program to create a Boolean expression that determines if a person is granted access to a government facility. Access is granted if at least two of the following three conditions are met:

- The person's fingerprint matches the stored fingerprint.
- The retinal scan matches the stored retinal data.
- The access card is valid.

Print `Access Granted` if the person is granted access, otherwise print `Access Denied`.

Test Case:

If `fingerprint_match = 1`, `retinal_match = 0`, `access_card_valid = 1`, the program should print `Access Granted`.

Question 4: Test Pass/Fail Condition

Write a C program to check if a student has passed a particular test. To pass, the student must have scored at least 70% on each of the three sections of the test, and the overall average score should be 75% or higher.

Test Case:

If `section1 = 70`, `section2 = 80`, `section3 = 75`, the program should print `Pass`.

Lab Submission Guidelines

- Compile a **small report including screenshots showing you work**
- **Rename the report** with **your IT number** and **covert it to a pdf** file (Example ITxxxxxxxxx.pdf)
- Upload the pdf file to the GitHub Classroom

